COURSE CODE & NAME: ECE1408- EMEBEDDED SYSTEM

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EX NO: 2

FORMAL DESIGN OF EMBEDDED CONTROLLER USING FSM – TRAFFIC LIGHT CONTROLLER

1. Aim

To design and simulate a traffic light controller using a Finite State Machine (FSM) to control traffic signals in a systematic sequence.

2. Software and Hardware Requirements

Software:

- Arduino IDE
- Proteus Design Suite

Hardware / Simulation Components:

- Arduino UNO
- Red LED indicators
- Yellow LED indicators
- Green LED indicators
- Resistors
- Virtual Terminal (if required)
- Connecting wires

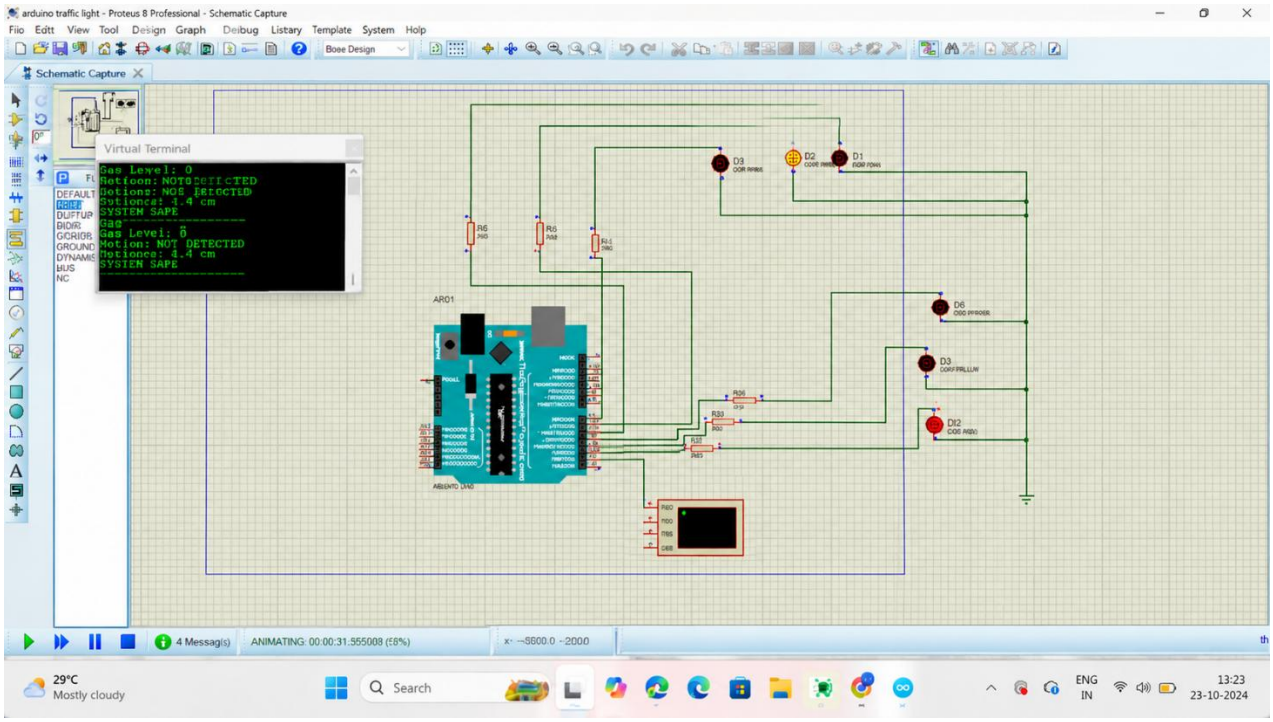
3. Program code:

```
// ------------------------------------------------
// Formal Design of Embedded Controller Using FSM
// Traffic Light Controller
// Arduino UNO + Proteus
// ------------------------------------------------
enum TrafficState
{
  STATE_NS_GREEN,
  STATE_NS_YELLOW,
  STATE_EW_GREEN,
  STATE_EW_YELLOW
};
TrafficState currentState = STATE_NS_GREEN;
// North-South traffic signal
const int NS_RED_PIN    = 2;
const int NS_YELLOW_PIN = 3;
const int NS_GREEN_PIN  = 4;
// East-West traffic signal
const int EW_RED_PIN    = 5;
const int EW_YELLOW_PIN = 6;
const int EW_GREEN_PIN  = 7;
void setup()
{
  pinMode(NS_RED_PIN, OUTPUT);
  pinMode(NS_YELLOW_PIN, OUTPUT);
  pinMode(NS_GREEN_PIN, OUTPUT);
  pinMode(EW_RED_PIN, OUTPUT);
  pinMode(EW_YELLOW_PIN, OUTPUT);
  pinMode(EW_GREEN_PIN, OUTPUT)
  Serial.begin(9600);
```

```
}
void allOff()
{
  digitalWrite(NS_RED_PIN, LOW);
  digitalWrite(NS_YELLOW_PIN, LOW);
  digitalWrite(NS_GREEN_PIN, LOW);
  digitalWrite(EW_RED_PIN, LOW);
  digitalWrite(EW_YELLOW_PIN, LOW);
  digitalWrite(EW_GREEN_PIN, LOW);
}
void enterState(TrafficState state)
{
  allOff();
  switch (state)
  {
    case STATE_NS_GREEN:
      digitalWrite(NS_GREEN_PIN, HIGH);
      digitalWrite(EW_RED_PIN, HIGH);
      Serial.println("STATE S0: NS GREEN - EW RED");
      break;
    case STATE_NS_YELLOW:
      digitalWrite(NS_YELLOW_PIN, HIGH);
      digitalWrite(EW_RED_PIN, HIGH);
      Serial.println("STATE S1: NS YELLOW - EW RED");
      break;
    case STATE_EW_GREEN:
      digitalWrite(NS_RED_PIN, HIGH);
      digitalWrite(EW_GREEN_PIN, HIGH);
      Serial.println("STATE S2: NS RED - EW GREEN");
      break;
    case STATE_EW_YELLOW:
```

```
      digitalWrite(NS_RED_PIN, HIGH);
      digitalWrite(EW_YELLOW_PIN, HIGH);
      Serial.println("STATE S3: NS RED - EW YELLOW");
      break;
  }
}
void loop()
{
  switch (currentState)
  {
    case STATE_NS_GREEN:
      enterState(STATE_NS_GREEN);
      delay(10000);
      currentState = STATE_NS_YELLOW;
      break;
    case STATE_NS_YELLOW:
      enterState(STATE_NS_YELLOW);
      delay(3000);
      currentState = STATE_EW_GREEN;
      break;
    case STATE_EW_GREEN:
      enterState(STATE_EW_GREEN);
      delay(10000);
      currentState = STATE_EW_YELLOW;
      break;
    case STATE_EW_YELLOW:
      enterState(STATE_EW_YELLOW);
      delay(3000);
      currentState = STATE_NS_GREEN;
      break;
  }
}
```

4.OUTPUT:



5. Result

Thus, the traffic light controller was designed and simulated using the Finite State Machine (FSM) concept. The controller operates the red, yellow, and green traffic signals in a predefined sequence to support systematic traffic management.